Kingdom of Cambodia

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EMR Clinical Data Submission Technical Specification

National Payment Certification Agency
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CONTENTS

1 OVERVIEW

This documentation provides a comprehensive reference guide for the Healthcare Application Programming Interface (API) data models. The API is designed to support the submission from multiple electronic medical records (EMR) that facilitates the management of patient information, clinical workflows, and healthcare services.

1.1 PURPOSE

The Healthcare API enables the digital transformation of healthcare services by providing standardized data structures for:

- Patient demographics and identification
- Clinical encounters and visits
- Medical observations and measurements
- Diagnostic procedures and results
- Clinical treatments and prescriptions
- Billing and payment processing

1.2 ARCHITECTURE

The API follows a modular architecture with interconnected components that represent different aspects of healthcare delivery:

1. **Base Components** - Learn about fundamental structures like Encounters and Observations that serve as building blocks for more complex operations
2. **Core Components** - Essential models such as Patient, Visit, Triage, Treatment, Invoices and other Clinical that form the core of the healthcare information system
3. **Specific Encounters** - Supporting structures categorized encounters that can easily help identify each case of a patient visit

1.3 DATA MODEL RELATIONSHIPS

The components are interconnected through reference codes:

- Each **Patient** can have multiple **Visits**
- Each **Visit** can include multiple **Encounters** (Outpatient, Inpatient, Emergency)
- Each **Encounter** can have associated **Observations**, **Diagnoses**, **Prescriptions**, and other clinical data

1.4 IMPLEMENTATION CONSIDERATIONS

The data models are designed to be interoperable with healthcare standards while addressing the specific needs of healthcare facilities in Cambodia, including support for local administrative divisions and healthcare facility structures.

1.5 Key Features

- Flexible and extensible components
- Flexible observation model that accommodates various data types
- Comprehensive medical history tracking
- Support for different encounter types (outpatient, inpatient, emergency)
- Integrated billing and payment systems
- Referral management between healthcare facilities
- Detailed clinical documentation with SOAP notes

2 API SPECIFICATIONS

This API provides endpoints to manage patient visits in the healthcare system. It allows submitting patient visits along with related data such as triage information, encounters, diagnoses, prescriptions, and more.

2.1 AUTHENTICATION

The API uses OAuth 2.0 with Password Client for authentication.

2.1.1 Obtaining Access Token

POST	/oauth/token

2.1.1.1 Request Body

```
{
  "grant_type": "client_credentials",
  "client_id": "your-client-id",
  "client_secret": "your-client-secret",
  "username ": "your-username",
  "password": "your-password",
}
```

2.1.1.2 Successful Response

```
{
  "token_type": "Bearer",
  "expires_in": 31536000,
  "access_token": "accessToken..."
  "refresh_token": "refreshToken..."
}
```

2.1.1.3 Error Response

```
{
  "error": "what_error",
  "message": "Error message here"
}
```

2.1.1.4 Using the Access Token

Include the access token in the `Authorization` header for all API requests:

```
Authorization: Bearer eyJ0eXAiOiJKV1QiLCJhbGciOiJSUzI1...
```

2.2 ENDPOINTS

2.2.1 Submit Patient Visit

POST	/api/external/v1/visits

This will submit a new patient visit record with all related information.

2.2.1.1 Request Headers

`Content-Type`	`application/json`
`X-System-Code`	`{system_code}`
`X-Health-Facility-Code`	`{health_facility_code}`
`Authorization`	`Bearer {access_token}`

- X-System-Code: a unique System code identifier provided by NPCA
- X-Health-Facility-Code: a unique Health Facility code identifier provided by NPCA
- Authorization: provide access token that received from authorization above

2.2.1.2 Body JSON Structure

The following is the overview of the JSON data structured to be submitted with the endpoint. Each are the components that will be defined later in the document.

```
{
  "visits": [{
    "patient": {},
    "triages": [{}],
    "outpatients": [{}],
    "inpatients": [{}],
    "emergencies": [{}],
    "surgeries": [{}],
    "progress_notes ": [{}],
    "soaps ": [{}],
    "vital_signs": [{}],
    "medical_histories": [{}],
    "physical_examinations": [{}],
    "laboratories": [{}],
    "imageries": [{}],
    "diagnoses": [{}],
    "prescriptions": [{}],
    "invoices": [{}],
    "referral": {}
  }]
}
```

Notes

- All datetime values should be in ISO 8601 format (YYYY-MM-DDThh:mm:ss.ssss+07:00)
- Patient code must be unique across the system
- Visit code and other codes are unique code that automatically generated by the EMR system
- If a field has no data available, set it to null instead of removing it, except blood glucose in vital signs
- Referral information must be included when patients are referred to or from other facilities

2.2.1.3 Successful Response


```
{
  "success": true,
  "http_code": 200,
  "message": "Visit submitted successfully",
}
```


2.2.1.4 Error Responses

2.2.1.4.1 400 Bad Request


```
{
  "success": false,
  "http_code": 400,
  "message": "Invalid request data",
  "errors": "Error Message"
}
```


2.2.1.4.2 401 Unauthorized


```
{
  "success": false,
  "http_code": 401,
  "message": "Unauthenticated",
  "errors": "Invalid or expired token"
}
```


2.2.1.4.3 403 Forbidden


```
{
  "success": false,
  "http_code": 403,
  "message": "Access denied",
  "errors": "You do not have permission to create visits"
}
```


2.2.1.4.4 500 Server Error


```
{
  "success": false,
  "http_code": 403,
  "message": "Internal server error",
```

```
  "errors": "An unexpected error occurred while processing your request"
}
```

3 BASE COMPONENTS

This section attempts to explain the base concept of each component that is used under the visit component.

3.1 ENCOUNTER

An Encounter represents a clinical interaction between a patient and healthcare provider during a visit.

3.1.1 Data Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
code	string	Unique identifier for the encounter
class	string	Class of encounter (“IPD”, “OPD”, or “EMERGENCY”)
name	string	What is the encounter (triage, ward names, sub-clinic …)
service_type	string	Type of the service provided (nullable)
started_at	datetime	Date and time encounter started
ended_at	datetime	Date and time encounter ended (nullable)
encountered_by	string	Name of healthcare provider
title	string	Whether the healthcare provider is doctor or nurse
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

3.1.2 Example

```
{
  "patient_code": "P25003877",
  "visit_code": "V420650",
  "code": "E317001",
  "class_name": "OPD",
  "name": "General Medicine"
  "service_type": "General Consultation",
  "started_at": "2025-03-17T09:35:00.000+07:00",
  "ended_at": "2025-03-17T10:15:00.000+07:00",
  "encountered_by": "Dr. Chhun Sovannarith"
  "title": "Doctor",
  "created_at": "2025-03-17T09:35:00.000+07:00",
  "updated_at": "2025-03-17T10:15:00.000+07:00"
}
```

3.2 OBSERVATIONS

Observations contain clinical data about a patient. They are designed to be flexible enough to capture virtually any type of patient information.

3.2.1 Data Model

Field	Type	Description
encounter_code	string	Reference to encounter unique code
observation_code	string	Reference to parent observation (nullable)
name	string	Name for the observation
category	string	Name for the category of the observation if applicable
value	various	Value of the observation (type depends on value_type)
value_type	string	Type of value (“boolean”, “string”, “text”, “integer”, “float”, “datetime”, “time”, “array”, “complex”)
value_unit	string	Unit of measurement (nullable)

3.2.2 Value Types

Observation structure needs to be flexible to allow any type of data store in EMR to be able to submit via API. To introduce flexibility, value type is necessary. It basically determines what type of data it is in the value field.

3.2.2.1 Boolean Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Smoker",
  "category": "Social History",
  "value": false,
  "value_type": "boolean",
  "value_unit": null
}
```

3.2.2.2 String Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Blood Type",
  "category": "Laboratory",
  "value": "O+",
  "value_type": "string",
  "value_unit": null
}
```

3.2.2.3 Text Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Clinical Notes",
  "category": "Assessment",
  "value": "Patient reports symptoms of malaria including periodic
fever, headache, and fatigue. Has history of travel to Mondulkiri
province two weeks ago.",
  "value_type": "text",
```

```
  "value_unit": null
}
```

3.2.2.4 Numeric Example


```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Blood Glucose",
  "category": "Vital Signs",
  "value": 110,
  "value_type": "integer",
  "value_unit": "mg/dL"
}
```


3.2.2.5 Datetime Example


```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Last Meal Time",
  "category": "Nutrition",
  "value": "2025-03-17T07:00:00.000+07:00",
  "value_type": "datetime",
  "value_unit": null
}
```


3.2.2.6 Time Example


```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Duration of Symptoms",
  "category": "History",
  "value": "48:00:00",
  "value_type": "time",
  "value_unit": "hh:mm:ss"
}
```


3.2.2.7 Array Example


```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Symptoms",
  "category": "History",
  "value": ["Fever", "Headache", "Fatigue", "Joint Pain"],
  "value_type": "array",
  "value_unit": null
}
```

3.2.2.8 Complex Example

```
{
  "encounter_code": "EN20250317001",
  "observation_code": null,
  "name": "Malaria Risk Assessment",
  "category": "Risk Assessment",
  "value": {
    "risk_level": "High",
    "travel_history": true,
    "prevention_measures": ["Bed nets", "Repellent"],
    "endemic_area": "Mondulkiri province"
  },
  "value_type": "complex",
  "value_unit": null
}
```

Note: conditions also have the same structure and values as observations. But for simplicity, the conditions in Medical History components are shortened and unify to make it easier for implementing it.

4 Core Components

4.1 Patient

The Patient object contains personal information and demographics for an individual under medical care.

4.1.1 Data Model

Field	Type	Description
code	string	Unique identifier for the patient in the system
surname	string	Patient's family name
name	string	Patient's given name
sex	string	Patient's biological sex ("F" or "M")
birthdate	date	Patient's date of birth
phone	string	Patient's contact number
nationality	string	Patient's nationality
disabilities	array of strings	List of disabilities if applicable
occupation	string	Patient's job or profession (nullable)
marital_status	string	Patient's marital status (nullable)
photos	array of strings	URLs to patient photos (nullable)
address	object	Patient's address information (see Address model)
identifications	array of objects	Patient identification documents (see Identification model)

death_date	datetime	Date and time of death if applicable
spid	string	Patient's Social Protection ID
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

4.1.2 Address Model

Field	Type	Description
province	object	Information about the province
province.code	string	Province code from NCDD standard
province.name	string	Province's name in Khmer
district	object	Information about the district
district.code	string	District code from NCDD standard
district.name	string	District name in Khmer
commune	object	Information about the commune
commune.code	string	Commune code from NCDD standard
commune.name	string	Commune name in Khmer
village	object	Information about the village
village.code	string	Village code from NCDD standard
village.name	string	Village name in Khmer
house_number	string	House number (nullable)
street_number	string	Street number (nullable)
location	string	Additional location details (nullable)

4.1.3 Identification Model

Field	Type	Description
patient_code	string	Reference to patient unique code
card_code	string	Identification number
card_type	string	Type of identification document

4.1.4 Example

```json
{
  "patient": {
    "code": "P25003877",
    "surname": "សៅ",
    "name": "សុវ៉ាន់",
    "sex": "M",
    "birthdate": "1980-05-15",
    "phone": "+855 12 345 678",
    "nationality": "Cambodian",
    "disabilities": ["Hearing impairment"],
    "occupation": "Teacher",
    "marital_status": "Married",
    "photos": ["https://hospital.example/photos/PT20250317001_1.jpg"],
    "address": {
      "province": {
```

```
        "code": "12",
        "name": "Phnom Penh"
      },
      "district": {
        "code": "1201",
        "name": "Chamkar Mon"
      },
      "commune": {
        "code": "120101",
        "name": "Tonle Bassac"
      },
      "village": {
        "code": "12010101",
        "name": "Phsar Deum Thkov"
      },
      "house_number": "45",
      "street_number": "302",
      "location": "Near Central Market"
    },
    "identifications": [
      {
        "patient_code": "P25003877",
        "card_code": "ID123456789",
        "card_type": "National ID"
      }
    ],
    "death_date": null,
    "spid": "250317001",
    "created_at": "2025-03-17T08:30:00.000+07:00",
    "updated_at": "2025-03-17T08:30:00.000+07:00"
  }
}
```


4.2 Visit

A Visit represents a patient's stay at a healthcare facility, from admission to discharge.

4.2.1 Data Model

Field	Type	Description
health_facility_code	string	Unique identifier for the healthcare facility, issued by NPCA
patient_code	string	Reference to patient unique code
code	string	Unique identifier for the visit
admission_type	string	How patient is admitted
discharge_type	string	How patient is discharged
visit_outcome	string	Outcome or condition of patient's illness
visit_type	string	Whether the visit is IPD or OPD
admitted_at	datetime	Date and time of admission
discharged_at	datetime	Date and time of discharge
followup_at	datetime	Date and time of scheduled follow-up

created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

4.2.2 Example


```
{
  "visits": [
    {
      "health_facility_code": "121020",
      "patient_code": "P25003877",
      "code": "V420650",
      "admission_type": "Walk-in",
      "discharge_type": "Authorized",
      "visit_outcome": "Improved",
      "visit_type": "OPD",
      "admitted_at": "2025-03-17T09:30:00.000+07:00",
      "discharged_at": "2025-03-17T10:45:00.000+07:00",
      "followup_at": "2025-03-24T09:00:00.000+07:00",
      "created_at": "2025-03-17T09:30:00.000+07:00",
      "updated_at": "2025-03-17T10:45:00.000+07:00"
    }
    ...
 ]
}
```


4.3 TRIAGE

Triage represents the initial assessment of a patient to determine treatment priority.

4.3.1 Data Model

Field	**Type**	**Description**
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code (nullable)
code	string	Unique identifier for the triage
chief_complaint	text	Primary reason for seeking care
height	number	Value of height of the patient in CM
weight	number	Value of weight of the patient in KG
recorded_at	datetime	Date and time triage was recorded
recorded_by	string	Name of healthcare provider who recorded the triage
title	string	Position of the healthcare provider

4.3.2 Example


```
{
  "triages": [{
    "visit_code": " V420650",
    "encounter_code": "P25003877",
    "code": "TR250317",
```

```
    "chief_complaint": "Periodic fever with chills for 3 days",
    "height": 170,
    "weight": 65,
    "recorded_at": "2025-03-17T09:32:00.000+07:00",
    "recorded_by": "Khun Srey Mom",
    "title": "Doctor",
  }]
}
```

4.4 Vital Signs

Vital Sign contains key physiological measurements of a patient. This object must contain all the vistal signs information such as temperature, pulse (heart rate), respiratory rate, systolic blood pressure, diastolic blood pressure, oxygen saturation, and an optional random rapid blood glucose test.

Vital Sign	Normal Range	Units
Temperature	36.5–37.3 °C	°C
Pulse (Heart Rate)	60–100 beats per minute	/mn
Respiratory Rate	12–20 breaths per minute	/mn
Blood Pressure	Systolic: 90–120 mmHg Diastolic: 60–80 mmHg	mmHg
Oxygen Saturation (SpO_2)	95–100%	%
Random Glucose Rapid Test	Random: < 140 mg/dL Fasting: 70–99 mg/dL	mg/dL

4.4.1 Data Model

Field	Type	Description
encounter_code	string	Reference to encounter unique code (such as triage, initial IPD, emergency, daily progress, procedure)
recorded_at	datetime	Date and time vital signs were recorded
recorded_by	string	Name of healthcare provider who recorded the vital signs
title	string	Doctor or nurse
observations	array of objects	Array of vital sign measurements

4.4.2 Observation in Vital Sign

Field	Type	Description
name	string	Name of the vital sign
value	numeric	Measured value

4.4.3 Example

```
{
  "vital_signs": [
    {
      "encounter_code": "E417001",
      "recorded_at": "2025-03-17T09:40:00.000+07:00",
      "recorded_by": "Khun Srey Mom",
      "title": "Doctor",
      "observations": [
```

```
        {
          "name": "Systolic Blood Pressure",
          "value": 135,
        },
        {
          "name": "Diastolic Blood Pressure",
          "value": 85,
        },
        {
          "name": "Heart Rate",
          "value": 92,
        },
        {
          "name": "Respiratory Rate",
          "value": 18,
        },
        {
          "name": "Temperature",
          "value": 38.7,
        },
        {
          "name": "Blood Oxygen",
          "value": 97,
        },
        {
          "name": "Random Blood Glucose",
          "value": 110,
        }
      ]
    }
  ]
}
```


4.5 Medical Histories

Medical history is a set of condition data that shows information clinically about the patient.

4.5.1 Components

Category	Section	Description
Current Illness	History of Present Illness	Detailed narrative of symptoms — onset, duration, severity, aggravating/relieving factors, associated symptoms
Current Illness	Current Medication	Current medications (name, dose, frequency), over-the-counter drugs, supplements
Health History	Past Medical History	Previous diagnoses (e.g., diabetes, hypertension), hospitalizations, chronic conditions
Health History	Past Surgical History	Details of past surgeries (type, date, complications if any)
Health History	Allergies	Known food, drug, environment reaction
Health History	Immunization	Vaccination status — especially for tetanus, flu, COVID-19, childhood vaccines

Women's Health	Obstetric & Gynecologic History	For female patients: menarche, menstruation pattern, pregnancies, contraception, menopause
Family & Lifestyle	Family History	Hereditary conditions in parents, siblings, grandparents. Example: heart disease, cancer
Family & Lifestyle	Social History	Lifestyle factors: tobacco use, alcohol, drug use, occupation, living situation, marital status

Note: the medical history detailed information must use these section names.

4.5.2 Data Model

Field	Type	Description
medical_history	array of conditions objects	Array of conditions (nullable)

4.5.3 Condition Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	text	Reference to encounter unique code
name	string	Name of the condition
value	array of string	Array of value in string of the condition

4.5.4 Example

```
{
  "medical_histories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": " Immunization History",
      "value": [
        "BCG",
        "OPV",
        "DTP",
        "Measles",
        "COVID-19"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Allergies",
      "value": [
        "Prahok",
        "Shrimp paste",
        "Penicillin"
      ]
    },
    {
```

```
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Past Surgical History",
      "value": [
        "Appendectomy Surgery"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Past Medical History",
      "value": [
        "Dengue fever (2020)",
        "Typhoid (2018)",
        "Cesarean section at Calmette Hospital (2019)"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Family History",
      "value": [
        "Mother with diabetes",
        "Father with hypertension and stroke at age 60"
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Current Illness History",
      "value": [
        "Patient reports high fever with chills occurring every 48
hours, accompanied by headache and joint pain. Symptoms started 3 days
ago after returning from a trip to Mondulkiri province. No prior history
of malaria. No medication taken yet."
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Current Medication",
      "value": [
        "Metformin 500mg",
        "Amlodipine 5mg"
      ]
    }
  ]
}
```

*Note: if value is empty or not available, the object must not be included

4.6 PHYSICAL EXAMINATIONS

Physical Examination contains clinical findings from examination of a patient.

4.6.1 Assessment Points

System/Section	Assessment Points
General Appearance	Consciousness (alert/drowsy), Distress, Nutrition, Posture, Gait, Hygiene, Pallor, Jaundice, Cyanosis, Edema
Glasgow Coma Scale	Eye opening, Verbal response, Motor response
Head	Scalp condition, Hair distribution, Lesions, Deformities
Eyes	Pupils (size, reaction), Conjunctiva (pallor), Sclera (icterus), Eye movements
Ears	Canal and Tympanic membrane, Discharge, Hearing
Nose	Nasal mucosa, Septum deviation, Discharge, Polyps
Throat/Mouth	Tonsils, Pharynx, Uvula, Oral mucosa, Tongue, Teeth
HEENT*	Information of Head, Eyes, Ears, Nose, Throat can be combined
Neck	Lymph nodes, Thyroid gland, Tracheal position, JVP
Cardiovascular System	Inspection (chest), Palpation (apex beat), Auscultation (S1/S2, murmurs), Peripheral pulses
Respiratory System	Chest symmetry, Expansion, Percussion (dullness), Breath sounds, Crepitations, Wheezing
Digestive System	Inspection (distension), Palpation (tenderness, organ size), Percussion, Bowel sounds
Genitourinary System	External genitalia (if applicable), Bladder palpation, Costovertebral angle tenderness, Discharge (if present)
Musculoskeletal System	Joint swelling, Deformity, Range of motion, Muscle tone/strength, Spine, Gait
Nervous System	Mental status, Cranial nerves, Motor/sensory function, Reflexes, Coordination
Skin	Rashes, Lesions, Color changes, Ulcers, Turgor, Temperature
Lymph Nodes	Cervical, Axillary, Inguinal – enlargement, tenderness, mobility

4.6.2 Data Model

Field	Type	Description
physical_examinations	array of object	Array of assessment points (nullable)

4.6.3 Physical Examination Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code
name	string	Name for the physical assessment point
value	various	Value of the assessment result
value_type	string	Type of Value
value_unit	string	Measurement unit represent the value

4.6.4 Example

```json
{
  "physical_examinations": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "General Appearance",
      "value": "Ill-appearing with intermittent shivering",
      "value_type": "text",
      "value_unit": null
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "name": "Skin",
      "value": "Warm, jaundiced, no rash",
      "value_type": "text",
      "value_unit": null
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E317001",
      "name": "Digestive System",
      "value": [
        {
          "inspection": "Flat, no visible masses"
        },
        {
          "palpation": "Soft, mild RUQ tenderness, spleen palpable 1cm below costal margin"
        },
        {
          "percussion": "No shifting dullness"
        },
        {
          "auscultation": "Normal bowel sounds"
        }
      ],
      "value_type": "complex",
      "value_unit": null
    }
  ]
}
```

4.7 LABORATORIES

4.7.1 Data Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code
request_code	string	Unique code represents the lab request
requested_at	datetime	Date and time laboratory were requested
requested_by	string	Name of doctor who requested the laboratory
title	string	Name of doctor
results	array of object	Array of laboratory objects

4.7.2 Result Model

Field	Type	Description
name	string	The item of lab result
category	string	Category that lab result fall under (e.g. biochemistry, hematology)
value	various	Measured value
value_type	string	Type of value
value_unit	string	Unit of measurement
reference_range	string	Normal range for the measurement
interpretation	string	Interpretation of the result
verified_at	datetime	Date and time of doctor who verified
verified_by	string	Name of healthcare provider who verified
recorded_at	datetime	Date and time of doctor who recorded
recorded_by	string	Name of healthcare provider who recorded

4.7.3 Example – Structured

```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-03-17T10:30:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "title": "Doctor",
      "results": [
        {
          "name": "Malaria Blood Smear",
          "category": "Parasitology",
          "value": "Positive for P. falciparum",
          "value_type": "string",
          "value_unit": null,
          "reference_range": "Negative",
          "interpretation": "Positive",
```

```
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        },
        {
          "name": "Parasitemia",
          "category": "Parasitology",
          "value": 2.5,
          "value_type": "float",
          "value_unit": "%",
          "reference_range": "0",
          "interpretation": "High",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        },
        {
          "name": "Hemoglobin",
          "category": "Hematology",
          "value": 11.2,
          "value_type": "float",
          "value_unit": "g/dL",
          "reference_range": "13.5-17.5",
          "interpretation": "Low",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        }
      ]
    }
  ]
}
```

4.7.4 Example – Complex

```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-04-22T09:15:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "results": [
        {
          "name": "Complete Blood Count",
          "category": "Hematology",
          "value": {
            "Hemoglobin": {
              "result": "11.2 g/dL",
```

```
        "reference": "13.5-17.5 g/dL",
        "interpretation": "Low"
      },
      "White Blood Cells": {
        "result": "4.5 x10^3/µL",
        "reference": "4.5-11.0 x10^3/µL",
        "interpretation": "Normal"
      },
      "Red Blood Cells": {
        "result": "4.1 x10^6/µL",
        "reference": "4.5-5.9 x10^6/µL",
        "interpretation": "Low"
      },
      "Platelets": {
        "result": "120 x10^3/µL",
        "reference": "150-450 x10^3/µL",
        "interpretation": "Low"
      },
      "Hematocrit": {
        "result": "34%",
        "reference": "41-50%",
        "interpretation": "Low"
      },
      "MCV": {
        "result": "83 fL",
        "reference": "80-96 fL",
        "interpretation": "Normal"
      },
      "MCH": {
        "result": "27.3 pg",
        "reference": "27-33 pg",
        "interpretation": "Normal"
      },
      "MCHC": {
        "result": "32.9 g/dL",
        "reference": "32-36 g/dL",
        "interpretation": "Normal"
      }
    },
    "value_type": "object",
    "value_unit": null,
    "reference_range": null,
    "interpretation": null,
    "verified_at": "2025-03-17T10:50:00.000+07:00",
    "verified_by": "Dr. Meas Chenda",
    "recorded_at": "2025-03-17T10:52:00.000+07:00",
    "recorded_by": "Meas Chandaravuth"
  },
  {
    "name": "Liver Function",
    "category": "Chemistry",
    "value": {
      "ALT": {
        "result": "45 U/L",
        "reference": "7-56 U/L",
```

```
          "interpretation": "Normal"
        },
        "AST": {
          "result": "42 U/L",
          "reference": "10-40 U/L",
          "interpretation": "High"
        },
        "ALP": {
          "result": "110 U/L",
          "reference": "44-147 U/L",
          "interpretation": "Normal"
        },
        "Total Bilirubin": {
          "result": "1.2 mg/dL",
          "reference": "0.1-1.2 mg/dL",
          "interpretation": "High-normal"
        },
        "Direct Bilirubin": {
          "result": "0.3 mg/dL",
          "reference": "0.0-0.3 mg/dL",
          "interpretation": "High-normal"
        },
        "Total Protein": {
          "result": "7.2 g/dL",
          "reference": "6.0-8.3 g/dL",
          "interpretation": "Normal"
        },
        "Albumin": {
          "result": "4.0 g/dL",
          "reference": "3.4-5.4 g/dL",
          "interpretation": "Normal"
        }
      },
      "value_type": "object",
      "value_unit": null,
      "reference_range": null,
      "interpretation": null,
      "verified_at": "2025-03-17T10:50:00.000+07:00",
      "verified_by": "Dr. Meas Chenda",
      "recorded_at": "2025-03-17T10:52:00.000+07:00",
      "recorded_by": "Meas Chandaravuth"
    }
   ]
  }
 ]
}
```

4.7.5 Example – File


```
{
  "laboratories": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "L7001",
      "requested_at": "2025-04-22T11:00:00.000+07:00",
      "requested_by": "Dr. Tep Veasna",
      "results": [
        {
          "name": "Blood Smear",
          "category": "Parasitology",
          "value":
["https://example.com/lab/EN20250422002/blood_smear_field1.jpg","https:/
/example.com/lab/EN20250422002/blood_smear_field2.jpg","https://example.
com/lab/EN20250422002/blood_smear_field3.jpg"
          ],
          "value_type": "array",
          "value_unit": null,
          "reference_range": null,
          "interpretation": "Positive for P. falciparum",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        },
        {
          "name": "Microbiology Culture",
          "category": "Microbiology",
          "value":
["https://example.com/lab/EN20250422002/culture_plate1.jpg","https://exa
mple.com/lab/EN20250422002/culture_plate2.jpg","https://example.com/lab/
EN20250422002/gram_stain.jpg","https://example.com/lab/EN20250422002/sen
sitivity_test.jpg"],
          "value_type": "array",
          "value_unit": null,
          "reference_range": null,
          "interpretation": "Positive for Salmonella typhi",
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        }
      ]
    }
  ]
}
```

4.8 Imageries

4.8.1 Data Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code
request_code	string	Unique code represents the lab request
requested_at	datetime	Date and time the imagery requested
requested_by	string	Name of healthcare provider who request the imagery
title	string	Name of dotor
results	array of object	Array of imagery objects

4.8.2 Result Model

Field	Type	Description
name	string	The item of imaging result
category	string	Category that imaging fall under (e.g. radiology, ...)
images	array of string	Measured value
result	string	Full result of the imagery
conclusion	string	Summary about the result
verified_at	datetime	Date and time that sample was verified
verified_by	string	Name of healthcare provider who verified the sample
recorded_at	datetime	Date and time results were recorded
recorded_by	String	Name of healthcare provider who recorded the result

4.8.3 Example

```
{
  "imageries": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "request_code": "IM2301",
      "requested_at": "2025-03-17T11:30:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "title": "title",
      "results": [
        {
          "name": "Brain Scan",
          "category": "CT Scan",
          "images": [
            "img.com/result2.jpg"
          ],
          "result": "string",
          "conclusion": null,
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
```

```
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        }
      ]
    },
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "requested_at": "2025-03-17T11:35:00.000+07:00",
      "requested_by": "Dr. Chhun Sovannarith",
      "tilte": "Doctor",
      "results": [
        {
          "name": "Adomen Ultrasound",
          "category": "Ultrasound",
          "images": [
            "img.com/result1.jpg"
          ],
          "result": "string",
          "conclusion": null,
          "verified_at": "2025-03-17T10:50:00.000+07:00",
          "verified_by": "Dr. Meas Chenda",
          "recorded_at": "2025-03-17T10:52:00.000+07:00",
          "recorded_by": "Meas Chandaravuth"
        }
      ]
    }
  ]
}
```


4.9 DIAGNOSIS

4.9.1 Diagnosis Types

Diagnosis Type	Description
Differential	A list of possible conditions that could explain the symptoms
In	The condition suspected when the patient is first admitted
Out	Final diagnosis after treatment and evaluation, recorded at discharge
Primary	The main reason for the current hospital visit or treatment
Secondary	Coexisting conditions that influence care or outcomes

4.9.2 Diagnosis Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code
diagnosis_type	string	Name of diagnosis type as above table
diagnosis_code	string	ICD-10 code for the diagnosis

diagnosis_name	string	Name of the diagnosis
diagnosis_description	string	Detailed description of the diagnosis
diagnosed_at	datetime	Date and time of diagnosis
diagnosed_by	string	Name of healthcare provider who made the diagnosis

4.9.3 Example


```
{
  "diagnosis": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E317001",
      "diagnosis_type": "Primary",
      "diagnosis_code": "B50",
      "diagnosis_name": "Plasmodium falciparum malaria",
      "diagnosis_description": "Plasmodium falciparum malaria with
cerebral complications",
      "diagnosed_at": "2025-03-17T10:05:00.000+07:00",
      "diagnosed_by": "Dr. Chhun Sovannarith"
    }
  ]
}
```


4.10 SOAPs

SOAPs are generally a set of texts that describe subjective, objective, assessment, evaluation and plan. In some cases, SOAPs are recorded in addition to all of the clinical details above or to summary the clinical details.

4.10.1 Data Model

Field	Type	Description
subjective	text	Patient's reported symptoms and feelings
objective	text	Provider's objective
assessment	text	Provider's assessment of the patient situation
evaluation	text	Provider's summary or evaluation regarding the assessment
plan	text	Plan for treatment or next steps

4.10.2 Example


```
{
  "soaps": [
    {

      "encounter_code": "EN20250317001"
      "subjective": "Patient reports fever has decreased after first
dose of medication. Still experiencing mild headache and fatigue. No
chills since this morning. Able to tolerate oral intake well.",
      "objective": "Temperature 37.8°C (down from 38.7°C). Heart rate
85/min. Blood pressure 125/80 mmHg. Skin less jaundice. Abdomen remains
```

```
soft with mild tenderness in RUQ.",
      "assessment": "Uncomplicated malaria showing initial response to
treatment. Fever is trending down. No signs of complications.",
      "plan": "Continue artemether-lumefantrine as prescribed. Encourage
oral hydration. Follow-up in clinic in 3 days. Return sooner if symptoms
worsen or unable to tolerate medication."
    }
  ]
}
```

4.11 Prescriptions

Prescription contains medication orders for a patient during an encounter.

4.11.1 Data Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code
code	string	Unique identifier for the prescription
prescribed_at	datetime	Date and time prescription was written
prescribed_by	string	Name of healthcare provider who wrote the prescription
title	string	Position whether doctor or nurse or else
medications	array of objects	Array of medications prescribed

4.11.2 Medication Model

Field	Type	Description
medicine_code	string	Unique identifier for the medication
medicine_name	string	Name of the medicine
strength	string	Strength of the medicine
form	string	Form of the medicine (tablet, syrup, etc.)
method	string	Method of administration
unit	string	Unit of measurement
morning	numeric	Morning dose
afternoon	numeric	Afternoon dose
evening	numeric	Evening dose
night	numeric	Night dose
days	integer	Duration of prescription in days
interval	string	Interval between doses
note	text	Additional notes or instructions

4.11.3 Example


```
{
  "prescriptions": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "code": "RX20250317001",
      "prescribed_at": "2025-03-17T10:10:00.000+07:00",
      "prescribed_by": "Dr. Chhun Sovannarith",
      "title": "Doctor",
      "medications": [
        {
          "medicine_code": "MED001",
          "medicine_name": "Artemether-Lumefantrine",
          "strength": "20/120mg",
          "form": "Tablet",
          "method": "Oral",
          "unit": "Tablet",
          "morning": 4,
          "afternoon": 0,
          "evening": 4,
          "night": 0,
          "days": 3,
          "interval": null,
          "note": "Take with food. Complete full course even if feeling
better."
        },
        {
          "medicine_code": "MED002",
          "medicine_name": "Paracetamol",
          "strength": "500mg",
          "form": "Tablet",
          "method": "Oral",
          "unit": "Tablet",
          "morning": null,
          "afternoon": null,
          "evening": null,
          "night": null,
          "days": 3,
          "interval": "8 hours",
          "note": "Take as needed for fever or pain."
        }
      ]
    }
  ]
}
```

4.12 Referral

Referral contains information about patient transfers between healthcare facilities.

4.12.1 Data Model

Field	Type	Description
referred_from	object	Information about the referring facility
referred_to	object	Information about the receiving facility

4.12.2 Detail Model

Field	Type	Description
code	string	Unique identifier for the referral
referral_number	string	Unique reference number for the referral
transportation	string	Mode of transportation
reason	string	Reason for referral
has_called	boolean	Whether the receiving facility was called in advance
caretaker_name	string	Name of the person accompanying the patient
caretaker_phone	string	Contact number for the caretaker
referred_by	string	Name of the referring provider
referred_by_phone	string	Contact number for the referring provider
referred_at	datetime	Date and time of referral
received_by	string	Name of the receiving provider
received_by_phone	string	Contact number for the receiving provider
received_at	datetime	Date and time patient was received
medications	string	Medications that the patient is taking

4.12.3 Example

```
{
  "referrals": {
    "referred_from": null,
    "referred_to": {
      "code": "REF20250317001",
      "referral_number": "HCPP/REF/2025/317",
      "transportation": "Ambulance",
      "reason": "Severe malaria with signs of organ dysfunction
requiring ICU care",
      "has_called": true,
      "caretaker_name": "Sok Channary",
      "caretaker_phone": "012345678",
      "referred_by": "Dr. Chhun Sovannarith",
      "referred_by_phone": "0987654321",
      "referred_at": "2025-03-17T11:30:00.000+07:00",
      "received_by": "Dr. Pich Vothy",
      "received_by": "0712345678",
      "received_at": "2025-03-17T12:15:00.000+07:00",
      "medications": "Artemether-Lumefantrine 20/120mg (1 dose given at
10:15), IV Artesunate 120mg initiated at 11:00"
    }
  }
}
```

4.13 INVOICES

Invoice records track payments for healthcare services and medications.

4.13.1 Data Model

Field	Type	Description
patient_code	string	Reference to patient unique code
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to encounter unique code
code	string	Invoice identifier
payment_type	string	Type of payment
invoice_date	datetime	Invoice date
total	number	Total amount
created_at	datetime	Creation timestamp
updated_at	datetime	Update timestamp
cashier	string	Name of cashier

4.13.2 Service Model

Field	Type	Description
invoice_code	string	Invoice identifier
service_code	string	Code identify the provided service
service_name	string	Name of the service provided
service_category	string	Category name of the service
price	number	Unit price
payment	number	Payment received from patient
paid	boolean	Payment status
discount_type	string	String whether in percentage or amount
discount	number	Discount amount or percentage

4.13.3 Medication Model

Field	Type	Description
invoice_code	string	Invoice identifier
medicine_code	string	Code identifies the provided service
medicine_name	string	Name of the service provided
quantity	number	Number of medicines provided
price	number	Unit price
payment	number	Payment received from patient
paid	boolean	Payment status
discount_type	string	String whether in percentage or amount
discount	number	Discount amount or percentage

4.13.4 Example


```json
{
  "invoices": [
    {
      "patient_code": "P25003877",
      "visit_code": "V420650",
      "encounter_code": "E417001",
      "code": "INV20250317001",
      "payment_type": "HEF",
      "invoice_date": "2025-03-17T11:00:00.000+07:00",
      "total": 120000,
      "created_at": "2025-03-17T11:00:00.000+07:00",
      "updated_at": "2025-03-17T11:00:00.000+07:00",
      "cashier": "Sok Kunthea",
      "services": [
        {
          "invoice_code": "INV20250317001",
          "service_code": "SRV001",
          "service_name": "Blood Smear for Malaria",
          "service_category": "Lab Test",
          "price": 15000,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        }
      ],
      "medications": [
        {
          "invoice_code": "INV20250317001",
          "medicine_code": "MED001",
          "medicine_name": "Artemether-Lumefantrine 20/120mg",
          "quantity": 24,
          "unit": "Tablet",
          "price": 1500,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        },
        {
          "invoice_code": "INV20250317001",
          "medicine_code": "MED002",
          "medicine_name": "Paracetamol 500mg",
          "quantity": 24,
          "unit": "Tablet",
          "price": 1000,
          "payment": 0,
          "paid": false,
          "discount_type": null,
          "discount": 0
        }
      ]
    }
  ]
}
```

5 Specific Encounters

5.1 Inpatient Admission

Inpatient Admission represents a hospital stay requiring overnight accommodation.

5.1.1 Data Model

Field	Type	Description
visit_code	string	Reference to visit unique code
code	string	Unique identifier for the operation
ward	string	Name of the ward or clinic or sub-clinic
service_type	string	Name of the service or purpose of inpatient (nullable)
started_at	datetime	Date and time operation started
ended_at	datetime	Date and time operation ended
encountered_by	string	Name of the healthcare provider
title	string	Position whether doctor or nurse or else
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

5.1.2 Example

```
{
  "inpatients": [
    {
      "visit_code": "V420650",
      "code": "E317001",
      "ward": "General IPD",
      "service_type": "Internal Medicine",
      "started_at": "2025-03-20T14:00:00.000+07:00",
      "ended_at": "2025-03-23T10:00:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title ": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00"
    }
  ]
}
```

5.2 Outpatient

Outpatient represents a hospital visit that doesn't require overnight accommodation.

5.2.1 Data Model

Field	Type	Description
visit_code	string	Reference to visit unique code
code	string	Unique identifier for the operation
ward	string	Name of the ward or clinic or sub-clinic
service_type	string	Name of the service or purpose of outpatient (nullable)
started_at	datetime	Date and time operation started
ended_at	datetime	Date and time operation ended
encountered_by	string	Name of the healthcare provider
title	string	Position whether doctor or nurse or else
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

5.2.2 Example

```
{
  "outpatients": [
    {
      "visit_code": "V420650",
      "code": "E317000",
      "ward": "General Inpatient",
      "service_type": "General Medicine",
      "started_at": "2025-03-17T09:35:00.000+07:00",
      "ended_at": "2025-03-17T10:15:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title ": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00",
      "created_at": "2025-03-17T09:35:00.000+07:00",
      "updated_at": "2025-03-17T10:15:00.000+07:00"
    }
  ]
}
```

5.3 EMERGENCY

Emergency represents urgent care provided in an emergency situation.

5.3.1 Data Model

Field	Type	Description
visit_code	string	Reference to visit unique code
code	string	Unique identifier for the operation
ward	string	Name of the ward or clinic or sub-clinic
service_type	string	Name of the service type or purpose (nullable)
started_at	datetime	Date and time encounter started
ended_at	datetime	Date and time encounter ended
emergency_note	text	Detailed notes about the emergency (nullable)
encountered_by	string	Name of the healthcare provider
title	string	Position whether doctor or nurse or else
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

5.3.2 Example

```
{
  "emergencies": [
    {
      "visit_code": "V420650",
      "code": "E317001",
      "ward": "Emergency",
      "service_type_name": "Emergency Medicine",
      "started_at": "2025-03-17T15:30:00.000+07:00",
      "ended_at": "2025-03-17T17:45:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title ": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00",
      "emergency_notes": "Patient brought by family after motorcycle
accident. GCS 14/15. No loss of consciousness. Skull X-ray negative for
fracture. CT not available. Multiple abrasions on right side. No
apparent fractures. Wound cleaning and dressing performed. Tetanus
prophylaxis given.",
      "created_at": "2025-03-17T15:30:00.000+07:00",
      "updated_at": "2025-03-17T17:45:00.000+07:00"
    }
  ]
}
```

5.4 Surgery

Surgery represents procedures that are done on patient in an operation theatre.

5.4.1 Data Model

Field	Type	Description
visit_code	string	Reference to visit unique code
encounter_code	string	Unique identifier for then encounter result in operation
code	string	Unique identifier for the operation
theater_name	string	Name of the operating theater
service_type	string	Name of the operation
started_at	datetime	Date and time operation started
ended_at	datetime	Date and time operation ended
reason	string	Reason for operating
anesthesia_type	string	Type of anesthesia used
procedure_notes	text	Detailed notes about the procedure
complications	array of strings	Any complications during the operation
specimens	array of strings	Specimens collected during the operation
blood_loss	string	Amount of blood loss during surgery (including unit)
surgeon_name	string	Name of the surgeon
anesthetist_name	string	Name of the anesthetist
assistant_names	array of strings	Names of assistants
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

5.4.2 Example


```
{
  "surgeries": [{
    "visit_code": "V420650",
    "encounter_code": "E401001",
    "code": "SUR20250401001",
    "theater_name": "Operating Theater 2",
    "service_type": "Laparoscopic Cholecystectomy",
    "started_at": "2025-04-01T08:30:00.000+07:00",
    "ended_at": "2025-04-01T10:15:00.000+07:00",
    "reason": "Symptomatic gallstones with recurrent biliary colic",
    "anesthesia_type": "General Endotracheal",
    "procedure_notes": "Patient positioned supine. Pneumoperitoneum
```

```
established with Veress needle. Four ports were placed. Gallbladder
dissected from liver bed with hook cautery. Cystic duct and artery
identified, clipped and divided. Gallbladder removed through umbilical
port. Hemostasis confirmed. No bile leak. Ports removed under direct
visualization. Wounds closed with 3-0 Vicryl and 4-0 Monocryl. Patient
tolerated procedure well.",
    "complications": [
      "Mild bleeding from liver bed, controlled with electrocautery"
    ],
    "specimens": [
      "Gallbladder with multiple stones",
      "Cystic fluid sample"
    ],

    "blood_loss": "5cl",
    "surgeon_name": "Dr. Sok Vichet",
    "anesthetist_name": "Dr. Meas Sokunthea",
    "assistant_names": [
      "Dr. Chhan Rathana" , "Nurse Van Bopha"
    ],
    "created_at": "2025-04-01T08:30:00.000+07:00",
    "updated_at": "2025-04-01T10:20:00.000+07:00"
  }]
}
```


5.5 Progress Notes

Daily Progress records the ongoing assessment and care of a patient, often using the SOAP format.

5.5.1 Data Model

Field	Type	Description
visit_code	string	Reference to visit unique code
encounter_code	string	Reference to parent encounter code
code	string	Unique identifier for the daily progress note
started_at	datetime	Date and time assessment started
ended_at	datetime	Date and time assessment ended
encountered_by	string	Name of the healthcare provider
title	string	Position whether doctor or nurse or else
created_at	datetime	Record creation timestamp
updated_at	datetime	Record last update timestamp

5.5.2 Example


```
{
  "progress_notes ": [
    {
      "visit_code": "VS20250317001",
      "encounter_code": "E417001",
      "code": "E317001",
      "started_at": "2025-03-17T16:00:00.000+07:00",
```

```
      "ended_at": "2025-03-17T16:15:00.000+07:00",
      "encountered_by": "Dr. Chhun Sovannarith",
      "title ": "Doctor",
      "created_at": "2025-03-20T14:00:00.000+07:00",
      "updated_at": "2025-03-23T10:00:00.000+07:00"
    }
  ]
}
```

Subject to be changed by NPCA

6 STANDARD DATA REFERENCE GLOSSARY

6.1 PATIENT DEMOGRAPHIC

This section defines the standard data for information relating to a patient

6.1.1 Sex

Value	Description
F	Patient who is a female from birth
M	Patient who is a male from birth

6.1.2 Marital Status

Value	Description
Married	Patient is legally married
Single	Patient has never been married
Widowed	Patient's spouse is decease
Divorced	Patient has been legally divorced and has not remarried
Separated	Patient is legally married but living apart from their spouse

6.2 VITAL SIGNS OBSERVATIONS

This section defines the standard vital signs recorded during patient assessment.

Concept	Description
Temperature	A float value representing the body's core thermal state.
Heart Rate	An integer that functions as a real-time counter for the heart's pump cycle, measured in beats per minute (bpm).
Respiratory Rate	An integer tracking the frequency of the body's primary resource-intake cycle (breathing).
Systolic Blood Pressure	An integer that measures the peak pressure on the circulatory network during a pump cycle.
Diastolic Blood Pressure	An integer that measures the baseline pressure when the pump is in its resting phase.
Blood Oxygen	A percentage integer that measures the saturation of the body's data carriers (hemoglobin) with oxygen.
Random Blood Glucose	An integer representing the amount of readily available fuel (glucose) in the system.

6.3 VISIT & ADMINISTRATIVE DATA

This section covers categorical data related to patient visits, admissions, and billing.

6.3.1 Admission

Value	Description
Walk-in	Patient comes to the facility without a prior appointment.
Appointment	Patient arrives for a scheduled visit.
Refer	Patient is referred by another facility or provider.

6.3.2 Visit Types

Value	Description
IPD	Patient stays at the hospital for one or more nights.
OPD	Patient receives care without being admitted overnight.

6.3.3 Outcome Types

Value	Description
Unchanged	No significant change in the patient’s condition.
Recovered	Patient’s condition has resolved.
Improved	Condition has shown improvement but is not fully resolved.
Hopeless	Patient's condition has no viable solution.
Home treatment	Patient can continue to receive treatment at home.

6.3.4 Discharge Types

Value	Description
Authorized	Patient is well enough to go home.
Unauthorized	Patient chose to leave before formal discharge.
Deceased	Patient passed away during the stay.
Hospitalization	Patient continues to receive treatment in the hospital.
Refer Out	Patient is sent to another hospital or care facility.
Absconded	Patient left without informing staff or completing a formal discharge.

6.3.5 Diagnosis Types

Value	Description
Differential	A list of possible conditions that could explain the symptoms.
In	The condition suspected when the patient is first admitted.
Out	The final diagnosis after treatment and evaluation, recorded at discharge.
Primary	The main reason for the current hospital visit or treatment.
Secondary	Coexisting conditions that influence care or outcomes.

6.3.6 Payment Types

Value	Description
HEF (Poor)	Patient is from poor families with payment is covered by the HEF
HEF (Informal Worker)	Patient is an informal worker with payment covered by the HEF
HEF (At Risk)	Patient is from family at risk with payment is covered by the HEF
NSSF Employee	Patient is an employee with payment is covered by the NSSF
NSSF Public Servant	Patient is public servant with payment is covered by the NSSF
NSSF Self-Employed	Patient is self-employed with payment is covered by the NSSF

6.3.7 Card Types

Value	Description
HEF	Health Equity Fund for Poor family
HEFR	Health Equity Fund for At Risk family
HEFIY	Health Equity Fund for Informal Worker with Yellow Card
HEFIV	Health Equity Fund for Informal Worker with Violet Card
NSSF	National Social Security Fund

NSSP	National Social Security Fund for Public servant
NSSPM	National Social Security Fund for Public servant's family Member
NSSE	National Social Security Fund for Employees
NSSEM	National social Security Fund for Employees' family Member
NSSFS	National Social Security Fund for Self-employed person
NSSFSM	National Social Security Fund for Self-employed person's family Member

6.4 MEDICAL HISTORY COMPONENTS

This section outlines the key areas to be documented when taking a patient's medical history.

- **History of Present Illness**: A detailed narrative of symptoms—onset, duration, severity, aggravating/relieving factors, and associated symptoms.
- **Past Medical History**: Previous diagnoses (e.g., diabetes, hypertension), surgeries, hospitalizations, and chronic conditions.
- **Past Surgical History**: Details of past operations including type, date, and any complications.
- **Current Medication**: A list of current medications (name, dose, frequency), over-the-counter drugs, and supplements.
- **Allergies**: Known drug, food, or environmental allergies and the type of reaction.
- **Family History**: Hereditary conditions in parents, siblings, and grandparents (e.g., heart disease, cancer).
- **Social History**: Lifestyle factors such as tobacco use, alcohol, drug use, occupation, living situation, and marital status.
- **Obstetric & Gynecologic History**: For female patients: menarche, menstruation pattern, pregnancies, contraception, and menopause.
- **Immunization History**: Vaccination status, especially for tetanus, flu, COVID-19, and common childhood vaccines.

6.5 PHYSICAL EXAMINATION COMPONENTS

- This section details the systems to be assessed during a physical examination and what to record for each.
- **General Appearance**: Consciousness (alert/drowsy), Distress, Nutrition, Posture, Gait, Hygiene, Pallor, Jaundice, Cyanosis, Edema.
- **HEENT**: Head, Eyes, Ears, Nose, Throat/Mouth.
- **Neck**: Lymph nodes, Thyroid gland, Tracheal position, Jugular Venous Pressure (JVP).
- **Cardiovascular System**: Inspection (chest), Palpation (apex beat), Auscultation (S1/S2, murmurs), Peripheral pulses.
- **Respiratory System**: Chest symmetry, Expansion, Percussion (dullness), Breath sounds, Crepitations, Wheezing.
- **Abdomen**: Inspection (distension), Palpation (tenderness, organ size), Percussion, Bowel sounds.
- **Musculoskeletal System**: Joint swelling, Deformity, Range of motion, Muscle tone/strength, Spine, Gait.
- **Nervous System**: Mental status, Cranial nerves, Motor/sensory function, Reflexes, Coordination.
- **Skin**: Rashes, Lesions, Color changes, Ulcers, Turgor, Temperature.
- **Lymph Nodes**: Cervical, Axillary, Inguinal – note any enlargement, tenderness, or changes in mobility.